

Static vs Contextual Embeddings for Lexical Ambiguity

A Word in Context Evaluation

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Introduction

“She sat on the **bank** of the river.” Here, *bank* means the land beside the water.

“She deposited money at the **bank**.” Here, *bank* means a financial institution.

The spelling is identical, but the meaning changes because of the surrounding words. This is **lexical ambiguity**.

We wanted to understand whether a computer model represents the same word differently when its context changes, and whether that difference helps it decide if the meaning is the same or different.

Inspiration

Our experiment follows the **Word in Context (WiC)** task introduced by Pilehvar and Camacho-Collados (2019). WiC avoids asking for a dictionary definition. It asks **whether the same word has the same meaning in two sentences**.

Camacho-Collados and Pilehvar (2018) describe a problem with traditional word representations. One stored vector can mix several meanings together. That motivated our comparison between a fixed representation and representations that use context.

Background

A **word embedding** is a list of numbers used to represent a word computationally. We compare two pretrained approaches called **GloVe** and **BERT**.

GloVe learns one stored vector for each word from patterns of words occurring together in large text collections. The vector for *bank* therefore stays the same in a river sentence and a money sentence. This is a **static embedding**.

BERT reads the surrounding sentence before creating the target-word representation. Its representation of *bank* can therefore change between those two sentences. This is a **contextual representation**.

The two meanings of *bank* above are **homonyms** because they are unrelated. Related meanings are called **polysemy**. For example, *paper* can mean writing material or a newspaper. The distinction is between unrelated and related meanings.

WiC does not ask the model to name a sense. It only asks whether the target word has the **same meaning** or a **different meaning** in the two sentences.

Hypothesis

We expected more context to make changes in meaning easier to recognise.

GloVe gives *bank* the same vector in both sentences, so it cannot see the difference between the river and money uses. Of the **638 validation pairs**, 319 have the same meaning and 319 have different meanings. Always choosing one answer would get half correct, so **50% is our simple baseline**.

We expected nearby words to help GloVe and the full sentence to help BERT more.

Methodology

The dataset

We used the **SuperGLUE Word in Context (WiC)** dataset throughout. Each example has **two sentences with the same marked word** and a label saying whether its meaning is the same.

A *bank* beside a river paired with a financial *bank* is labelled **different meaning**. Two financial uses of *bank* would be labelled **same meaning**.

WiC has **5,428 labelled training pairs** and **638 labelled validation pairs**. We chose method settings on training data, then measured performance on validation.

The **1,400 test pairs** have no public labels, so we report validation scores.

The comparison

We used **GloVe and BERT in three conditions**, always on the same data.

1. GloVe target word only. We use its stored vector with 300 numbers. The same word gets the same vector in both sentences.

2. GloVe with nearby words. We average the vectors of up to **two words before and after the target**, leaving out the target.

“The **muddy river** *bank* **flooded after** rain.”
For *bank*, the context words are **muddy, river, flooded, after**.

3. BERT using the full sentence. We use a frozen pretrained BERT base model. We combine the target representation from its **last four layers** without training it further on WiC.

From vectors to an answer

For each sentence pair, a method gives us two target representations. We compare them with **cosine similarity**. A higher score means the representations are more alike. A lower score means they are more different.

Using training data, we select a cutoff for each condition. Scores above the cutoff predict **same meaning**, while scores below it predict **different meaning**. Our main score is **accuracy**, the percentage of the 638 validation pairs answered correctly.

Results

All three conditions were evaluated on the same **638 validation pairs**.

GloVe using the target word alone answered **319 pairs correctly (50.0%)**. Adding nearby GloVe words increased this to **354 pairs (55.5%)**. BERT performed best with **428 correct pairs (67.1%)**.

Validation accuracy

GloVe target only		50.0%	(319/638)
GloVe nearby words		55.5%	(354/638)
BERT in full sentence		67.1%	(428/638)

+11.6 percentage points

Compared with GloVe using nearby words, BERT answered **74 additional sentence pairs correctly**.

We repeated that comparison on **1,000 resampled versions** of the validation set. The estimated 95% interval for BERT's advantage was **6.6 to 16.9 percentage points**, so the advantage remained positive throughout this interval.

Looking pair by pair, both stronger methods were correct on **250 pairs**. BERT alone was correct on **178**, GloVe with nearby words alone on **104**, and both were wrong on **106**. BERT was stronger on this benchmark, but neither method solved every example.

Conclusion

Our results support the idea that **context helps a model recognise a word's meaning**. The fixed GloVe vector could not distinguish the two uses of a word. Nearby words helped a little, and BERT's representation of the full sentence did better.

BERT still missed **210 of 638 pairs**. We found a clear improvement, not a complete solution. Building and checking this comparison ourselves made the difference between static and contextual embeddings much easier to see.